title: Compass / Attitude Sensor description: The Slocum compass and attitude sensor (True North TNT Revolution / legacy TCM3, the attitude_rev device): what it measures, magnetic variation, when and how to calibrate it on shore with RevolutionTest, the in-situ at-sea calibration, the post-cal compass check, and field troubleshooting for heading errors and interference.

Compass / Attitude Sensor

A precision navigation **compass and attitude sensor** continuously measures the glider's **heading, pitch, and roll**. These are the inputs the glider uses to **dead-reckon** its position underwater between GPS fixes, so a compass that is off by a few degrees turns directly into navigation error — the glider thinks it is flying one way while the water carries it another.

On modern G2/G3/G3S gliders the device is a **True North Technologies "TNT" Revolution** attitude module, addressed in the flight code as the **`attitude_rev`** device. Very early gliders (roughly serial numbers below ~200, G1-era) instead carried a **PNI TCM3** compass with its own `tcm3 cal` routine. This page focuses on the TNT/ `attitude_rev` device that the vast majority of the fleet uses.

Source

Paraphrased from the *Slocum Glider Operators Manual* (Rev. 1, "Attitude Sensor") and the *Slocum G3 Glider Maintenance Manual* (Rev. A, "Calibrating the True North Compass"), `masterdata.txt`, the Teledyne Webb Research user forum, and the UG2 community Slack (including in-situ calibration user notes contributed by NorGliders). For the calibration software itself, refer to True North Technologies' *Revolution Compass User's Guide*. This is a condensed field reference — always defer to the official Teledyne documentation and the `readme.txt` shipped with the calibration tool for your specific glider.